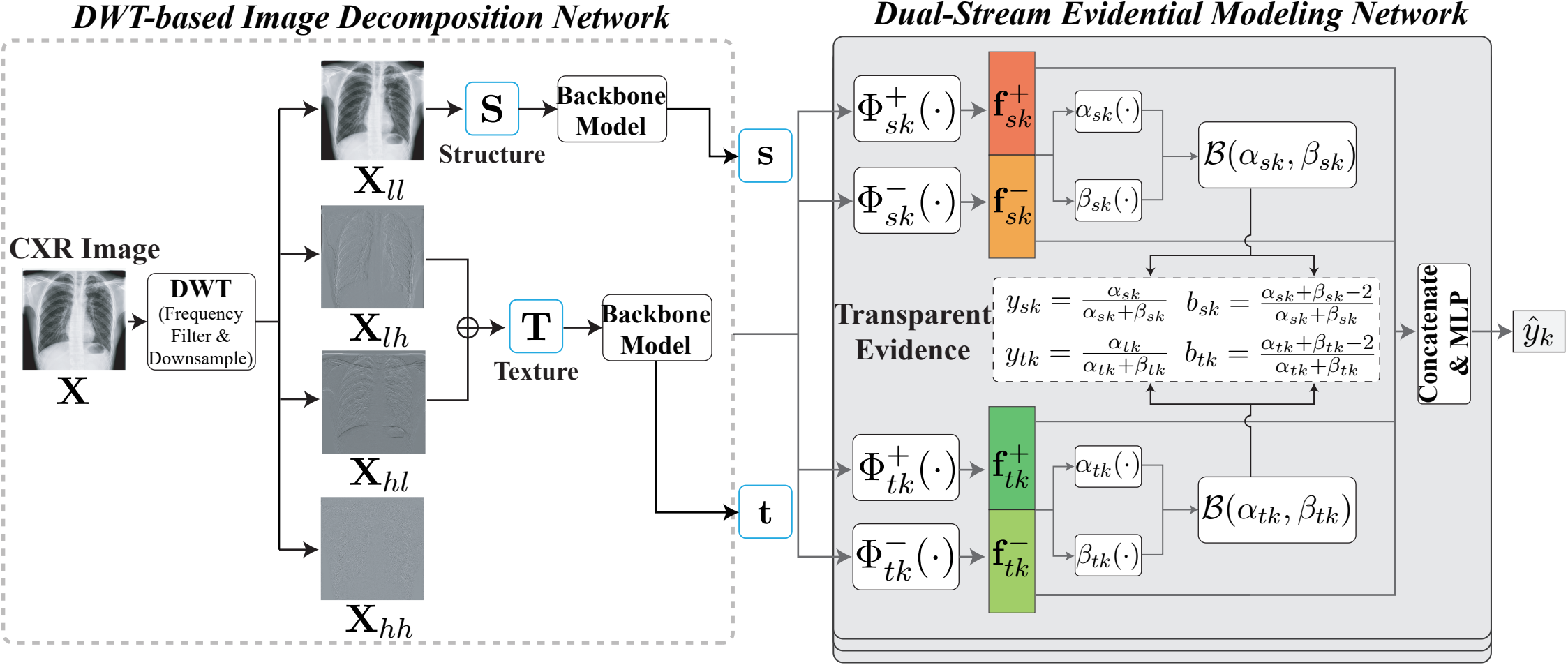

DWT-based Image Decomposition Network
Dual-Stream Evidential Modeling Network
CXR Image
X
DWT
(Frequency Filter & Downsample)
$\mathbf{X}_{ll}$
$\mathbf{X}_{lh}$
$\mathbf{X}_{hl}$
$\mathbf{X}_{hh}$
S
Structure
T
Texture
Backbone Model
Backbone Model
s
t
$\Phi^{+}_{sk}(\cdot)$
$\Phi^{-}_{sk}(\cdot)$
$\mathbf{f}^{+}_{sk}$
$\mathbf{f}^{-}_{sk}$
$\alpha_{sk}(\cdot)$
$\beta_{sk}(\cdot)$
$\mathcal{B}(\alpha_{sk}, \beta_{sk})$
Transparent Evidence
$y_{sk} = \frac{\alpha_{sk}}{\alpha_{sk}+\beta_{sk}}$ $b_{sk} = \frac{\alpha_{sk}+\beta_{sk}-2}{\alpha_{sk}+\beta_{sk}}$
$y_{tk} = \frac{\alpha_{tk}}{\alpha_{tk}+\beta_{tk}}$ $b_{tk} = \frac{\alpha_{tk}+\beta_{tk}-2}{\alpha_{tk}+\beta_{tk}}$
$\Phi^{+}_{tk}(\cdot)$
$\Phi^{-}_{tk}(\cdot)$
$\mathbf{f}^{+}_{tk}$
$\mathbf{f}^{-}_{tk}$
$\alpha_{tk}(\cdot)$
$\beta_{tk}(\cdot)$
$\mathcal{B}(\alpha_{tk}, \beta_{tk})$
Concatenate & MLP
$\hat{y}_k$